AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM359
Systems Penetration Testing

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 21/April/2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2h

Instructions:

- Total Marks: **60**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [10 marks]

For each question fill in the blank space with an appropriate word, acronym, name or phrase. For each blank space you must give only one answer. However, there may be more than one correct answer. Each question is worth 1 mark.

1) Ethical hacking is conducted to improve the organization's overall _____ posture.
2) Passive reconnaissance gathers information without directly interacting with the _____ system.
3) During a SYN scan, if a port is open, the attacker typically receives a SYN/_____ response.
4) DNS zone transfer can reveal the complete _____ structure of a domain.
5) NetBIOS enumeration may disclose shared _____ on a Windows machine.
6) SNMP stores management information inside a structured database called _____.
7) Vulnerabilities may arise due to insecure system _____.
8) Zero-day attacks exploit vulnerabilities before a _____ becomes available.
9) Password hashes in Windows systems are stored in the _____ database.
10) A rootkit is designed to conceal malicious activities and maintain unauthorized _____.

Question 2 [10 Marks]

A. Explain the Six (6) phases of Ethical Hacking methodology. (6 Marks)
B. Briefly explain DNS Footprinting and why it is important. (4 Marks)

Question 3 [10 Marks]

Consider the following command:

hydra -l admin -P passwords.txt 192.168.1.30 ssh

Analyze the command and answer the following:

A. Identify the type of attack being performed? (1 Marks)
B. State what the argument " -l admin " specify? (2 Marks)
C. What is the purpose of -P passwords.txt? (2 Marks)
D. Which service is being targeted? (2 Marks)
E. Mention two defensive measures against this attack. (3 Marks)

Question 4 [10 Marks]

A. Define Vulnerability Assessment. List and explain at least **Three (3)** types of information obtained from a vulnerability scanner. (5 Marks)
B. Explain Vulnerability Research and state how vulnerabilities are classified. (5 Marks)

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Question 5 [10 Marks]

1. An organization hires a security professional to simulate real-world attacks without causing damage and within legal boundaries. Which statement best describes this activity?
   A. It is an unauthorized attempt to bypass security controls
   B. It is a malicious attack intended to disrupt operations
   C. It is a structured security assessment conducted with permission
   D. It is cyber espionage performed under supervision
2. A security analyst gathers employee email addresses from social media and public websites without interacting with the target network. This activity is classified as:
   A. Active reconnaissance
   B. Passive reconnaissance
   C. Enumeration
   D. Vulnerability exploitation
3. Which statement best reflects the purpose of vulnerability assessment?
   A. It guarantees that no attacks will occur
   B. It replaces penetration testing
   C. It recognizes, measures, and classifies security vulnerabilities in a computer system, network, and communication channels
   D. It blocks all malicious traffic automatically
4. Which of the following best describes a threat actor motivated primarily by ideology rather than financial gain?
   A. Script Kiddie
   B. Hacktivist
   C. Cyber Criminal
   D. Insider
5. Which sequence correctly represents the TCP three-way handshake process?
   A. SYN → FIN → ACK
   B. SYN → RST → ACK
   C. ACK → SYN → RST
   D. SYN → SYN-ACK → ACK

Question 6 [10 marks]

Answer the following:

A. Explain the difference between passive and active footprinting. Discuss advantages and risks of each. (6 Marks)
B. Explain the TCP three-way handshake and discuss how attackers use it during network scanning.(4 Marks)

End of the questions.

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